Assignment for Week 8

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Package Loading

```
library(ggplot2)
library(dplyr, warn.conflicts = F)
library(readr)
```

Assignment

1) 5.1 (p 284)
2) 5.8 (p 290)
3) 9.20 (p 518)
4) 9.22 (a) (b) (p 519)
5) 9.26 (p 519)
6) 9.27 (p 519)
7) 9.30 (p 520)
8) Suppose $\bar{X}$ is the mean of 100 observations from a popualtion with mean μ and variance σ^2. Find limits between which $\bar{X} - \mu$ will lie with a probability of at least .85 using the Central Limit Theorem.
9) Consider the function $f(x, y)$ defined by the following table

Table 1: $f(x, y)$

	y= 1	2	3
x= 1	.1	0	.1
2	0	.2	.2
3	.1	.2	.1

(a) Verify that $f(x, y)$ is a PDF.
(b) Find the marginal probability functions of X and Y.
(c) Compute E(X) and V(X).
(d) Compute the covariance between X and Y.
(e) Find $P(X < Y)$.
(f) Are X and Y independent? Justify your answer.

R assignments from Chapters 9 and 10:

10) Plot the number of bombs dropped in each contemporary European country during World War I.

11) Estimate the location (coordinates) of the center of gravity of bombs dropped during World War I using the Thor dataset. Each row of the Thor dataset describes many bombs being dropped. How does that average move over the course of the war? Estimate its location for 1, 3, and 6 month rolling averages.

```r
URLchunkA <- "https://raw.githubusercontent.com/washingtonpost/"
URLchunkB <- "data-police-shootings/master/fatal-police-shootings-data.csv"
shootings <- read_csv(paste0(URLchunkA, URLchunkB))
```

```
## Parsed with column specification:
## cols(
##   id = col_integer(),
##   name = col_character(),
##   date = col_date(format = ""),
##   manner_of_death = col_character(),
##   armed = col_character(),
##   age = col_integer(),
##   gender = col_character(),
##   race = col_character(),
##   city = col_character(),
##   state = col_character(),
##   signs_of_mental_illness = col_character(),
##   threat_level = col_character(),
##   flee = col_character(),
##   body_camera = col_character()
## )
```

12) This question asks us to use ggplot to display the timing of police shootings data above.

a) First, use the `lubridate` package to round dates to the nearest month. Use the `round_date` function.

b) Display these data with a histogram with 10 bins, 20 bins, 30 and 35 bins. Do the same with a density plot with varying adjustment equal to .1, .2, and .8. Describe what is happening.

c) Produce a table which displays the count of white people that were shot and the presence of a body camera.

d) What would the expected counts be if these two variables (being white and body camera presence) were independent?

e) What is the difference between the observed counts and the expected counts?

f) Square the answers from e), divide by the expected count, and take the sum. This is called the chi-square statistical test of independence. It is distributed chi-square with `2*2 -1 =3` degrees of freedom.

g) Use the cdf function for the chi-square to describe the probability of observing the result from f) were these two variables independent.

h) Replicate the above steps with the chisq.test command.